The use of elliptic-curve isogenies in post-quantum cryptography: an analysis of the SIDH and CSIDH algorithms for secure key exchange

Kostiantyn Boiar

Introduction

The aim of this work is to study and perform a comparative analysis of post-quantum cryptographic key-agreement algorithms based on isogenic transformations of elliptic curves.

The scientific novelty of the work lies in the investigation of modern post-quantum key-agreement algorithms based on elliptic-curve isogenies, which in the future may solve the problem of ensuring the resistance of cryptographic protocols to attacks by quantum computers.

The practical value of the results lies in the development of a software product that can be used in a wide variety of systems where secure key exchange between users is of primary importance — for example, for two-factor user authentication.

Introduction

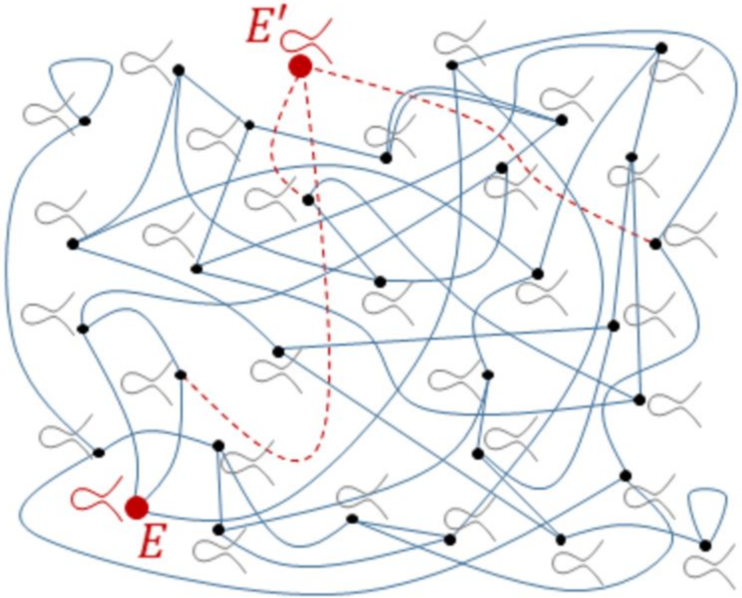
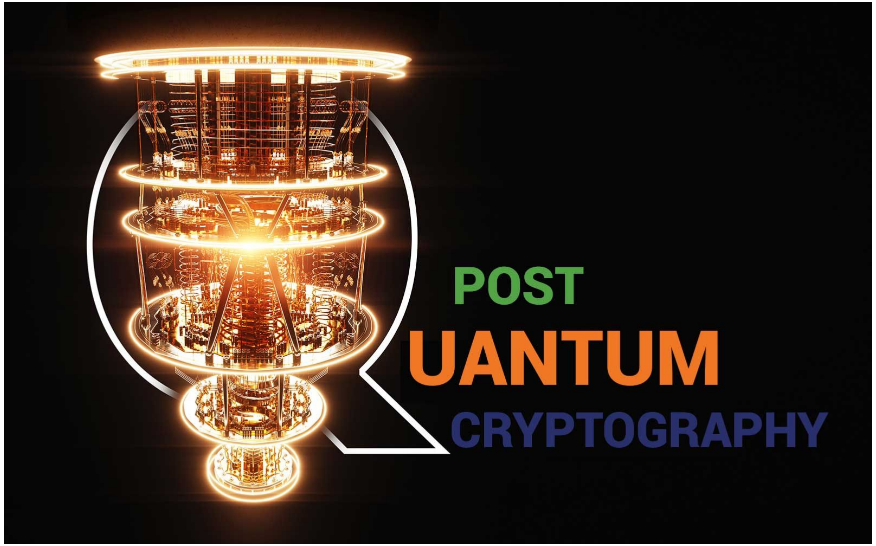


Figure 1 – Principle of an isogenous transformation [2]



Mathematical Foundations of Cryptography Based on Elliptic-Curve Isogenies

- Quadratic extensions of fields $F_p \equiv 3 \pmod 4$
- Most convenient representation $F_{p^2} = F_p(i)$ where $i^2 + 1 = 0;$
- Form of elements $u + vi$, where $u, v \in F_p$

Properties of the supersingular eliptic curve

1. $\#E(\mathbb{F}_q) = q + 1 - t$ Where $t \equiv 0 \pmod{p}$

2. $E[p] = \infty$

Finding supersingular j-invariants of curves

1. $\lfloor p/12 \rfloor + z$, where $z \in 0, 1, 2$. depends on $p \bmod 12$
2. $s_p^2 = \lfloor 241/12 \rfloor + 0 = 20$
3. EC in Weierstrass form: $y^2 = x^3 + ax + b$
4. j-invariant: $j(E) = 1728 \frac{4a^3}{4a^3 + 27b^3}$

Finding the kernel of an isogeny

- Take an elliptic curve in Montgomery form $M_{A,B} : By^2 = x^3 + Ax^2 + x$

- Apply the multiplication operation by x (consider multiplication by 2) $[2] : E_a \to E_a, x \to \dfrac{(x^2-1)^2}{4x(x^2+ax+1)}$

- Find exceptional points $a^2 + \alpha x + 1 = 0$, where $(0,0), (\alpha,0), (1/\alpha,0)$

- Form a subgroup of points on the elliptic curve $\ker([2]) \cong \mathbb{Z}_2 \times \mathbb{Z}_2$

Finding an isogeny of an elliptic curve

- We set $G = (0,0),(\alpha,0),(1/\alpha,0))$
- We obtain the mapping using Vélu's formulas $\phi : E_a \to E_{a'}, x \to \dfrac{x(\alpha x - 1)}{x - \alpha},$

where $a' = 2(1 - 2\alpha^2)$

- We choose an arbitrary parameter a for E_a , let $a = 93i + 143$
- Having obtained the kernel value $\alpha = 15i + 26$, we calculate the parameter a'

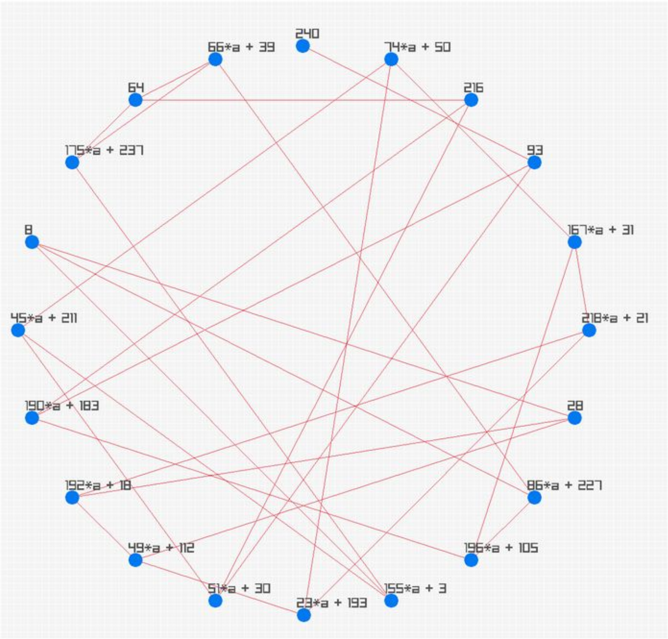
$$a' = 2(1 - 2 * (15 * i + 26)^2) = 1802 + 3120i$$

- Now, having obtained the value a' , we get a curve of the form

$$E'_a : y^2 = x^3 + (1802 + 3120i)x^2 + x, j(E'_a) = 86i + 227$$

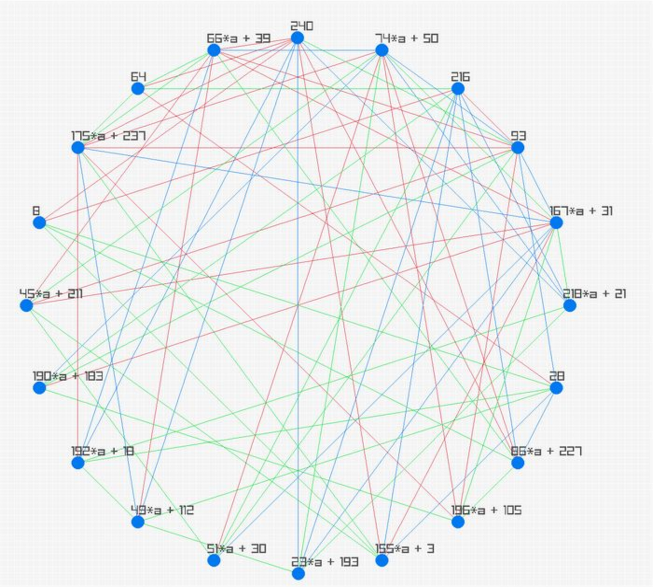
Visualization of the isogeny graph for a given degree

For each prime p we work with the set of all j-invariants in $\mathbb{F}_{p^2}$:



Visualization of the isogeny graph for a given degree

- This can be done for any value $ker([l]) \cong \mathbb{Z}_l \times \mathbb{Z}_l$
- In our example, we will visualize the isogeny graph for values $l = 2, 3, 5$



Example of using isogenies based on the SIDH algorithm. Public key generation

- *Input:* Elliptic curve, initial EC point, degree of isogenies.
- *Output:* Public key (path through graph), two resulting EC points

Public Key Generation

Algorithm 1 Public Key Generation

Input: Start
Choose a_0
Generate E_{a_0}, a_0, j_0
Choose P, Q on EC E_{a_0}
$n \leftarrow 0$
while $n <$ number of invariants **do**
    Choose k
    Find $k_Q = k \cdot Q$
    Find $S_a = k_Q + P$
    Find isogeny $E_{a_{n-1}} \rightarrow E_{a_n}$
    Find j-invariant of isogeny
    $n \leftarrow n + 1$
end while
Output: Path through graph, points P, Q

Private key generation

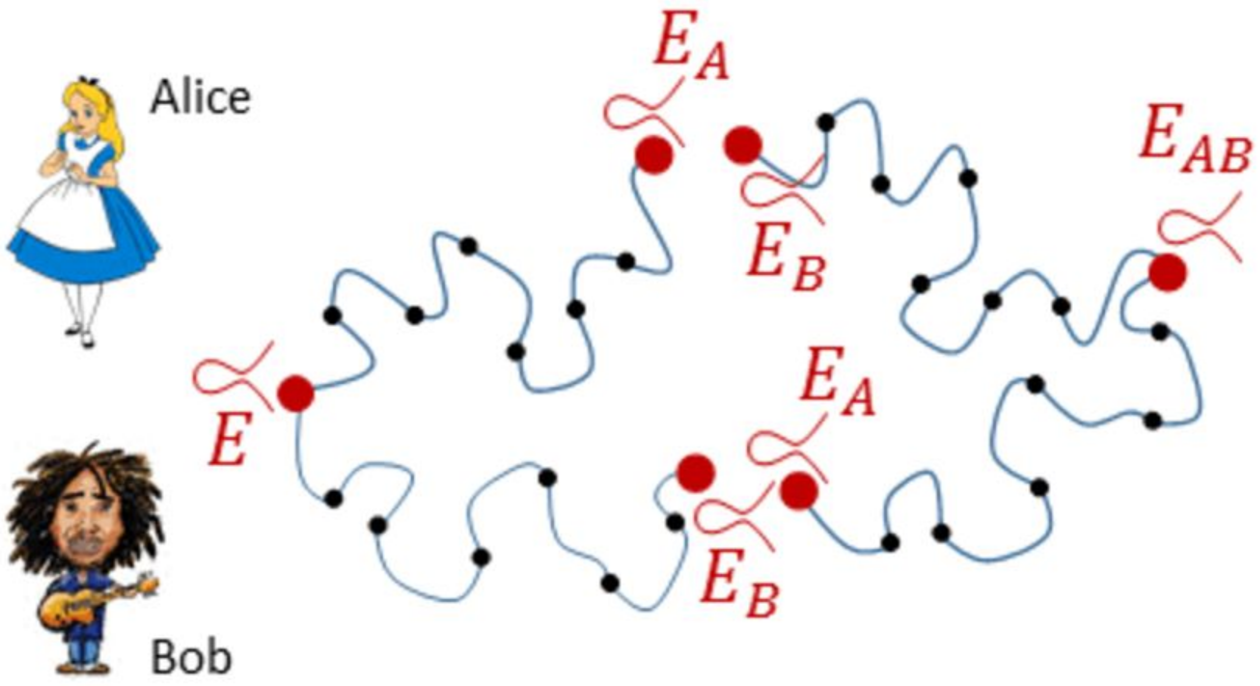
- *Input:* Path through graph of public key, initial EC point
- *Output:* Path through graph (private key)

Private Key Generation

Algorithm 2 Private Key Generation

Input: Start
Choose a_0
Calculate E_{a_0}, a_0, j_0
Choose P, Q on EC E_{a_0}
$n \leftarrow 0$
while $n <$ number of invariants **do**
    Choose k
    Find $k_Q = k \cdot Q$
    Find $S_a = k_Q + P$
    Find isogeny $E_{a_{n-1}} \rightarrow E_{a_n}$
    Find j-invariant of isogeny
    $n \leftarrow n + 1$
end while
if paths meet at one point **then**
    Output: Key matched
else
    Output: Key did not match
end if

Private Key Generation



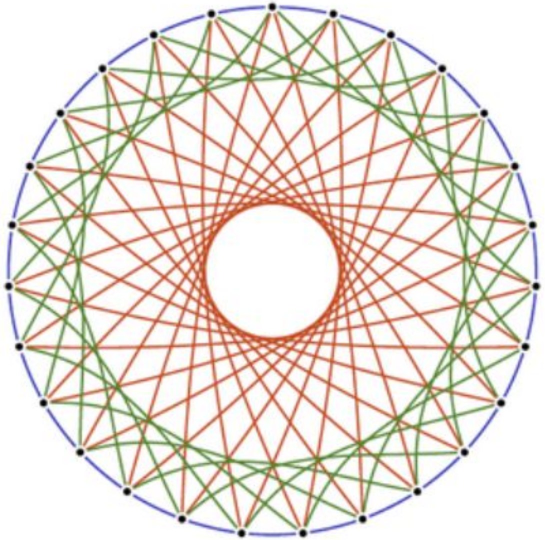
Private Key Generation Visualisation [2]

Comparison of CSIDH and SIDH algorithms

Table 1: Comparison of CSIDH and SIDH algorithms

SIDH	**CSIDH**
Extended Galois field	Prime Galois field
Uses the ring of endomorphisms	Based on commutative operations similar to operations over ideal classes
Movement through the graph, where from each vertex exits a number of edges equal to the degree of isogeny	Movement through the graph is performed "left" and "right" (along pairs of twisted curves) through subgraphs that correspond to l_i-isogenies
Slower	Faster

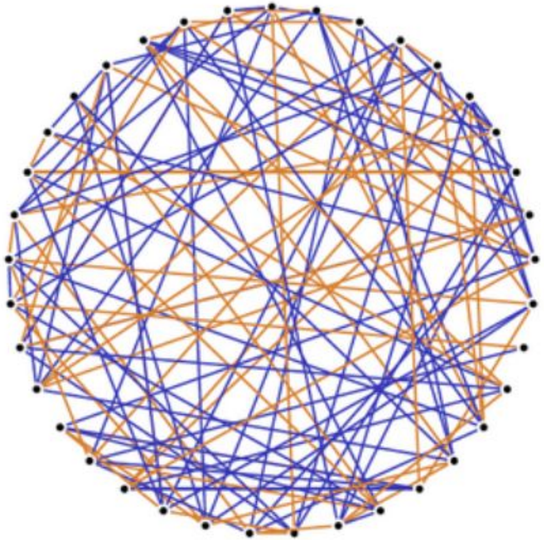
Comparison of isogeny graphs in CSIDH and SIDH



$q = p$

CSIDH ['siː,saɪd]

https://csidh.isogeny.org



$q = p^2$

SIDH

https://sike.org

Twisting in the CSIDH algorithm

- A twist E'/k of an elliptic curve E/k is called a curve E' that is isomorphic to E over $\overline{k}$
- For $E : y^2 = x^3 + Ax^2 + x$ the curve $E' : -y^2 = x^3 + Ax^2 + x$ is isomorphic through the mapping $(x, y) \to (x, \sqrt{-1}y)$
- Since E' does not have the standard Weierstrass form, it can be transformed into the form $E'' : y^2 = x^3 - Ax^2 + x$ using the isomorphism $(x, y) \to (-x, y)$ over $\mathbb{F}_p$

Twisting in the CSIDH algorithm

Each value $x \in \mathbb{F}_p$ satisfies one of the following cases:

1. If $x^3 + Ax^2 + x$ is a square in $\mathbb{F}_p$, then there are two

   points $(x, \pm\sqrt{x^3 + Ax^2 + x}) \in E(\mathbb{F}_p)$

2. If $x^3 + Ax^2 + x$ is not a square in $\mathbb{F}_p$, then there are two

   points $(x, \pm\sqrt{-(x^3 + Ax^2 + x)}) \in E'(\mathbb{F}_p)$

3. If $x^3 + Ax^2 + x = 0$, then the point $(x, 0)$ belongs to both $E'(\mathbb{F}_p)$ and $E(\mathbb{F}_p)$

Public parameters of CSIDH

- **CSIDH parameters**
  - $p = 4 \cdot l_1 \cdot l_2 \cdot ... \cdot l_n - 1$
  - Supersingular elliptic curve over prime field $\mathbb{F}_p$
- Isogenies defined over field $\mathbb{F}_p$ are used, which correspond to the chosen prime numbers l_i

Comparison of key agreement

Protocol	Key Size (bits)	Key Gen time	Secret Derivation Time	Quantum Security	Common Uses
ECDH (Curve25519)	256	< 1 ms	< 1 ms	No	TLS, WireGuard
ECDH(secp256k1)	256	< 1 ms	< 1 ms	No	Bitcoin, Ethereum
SIDH	512	50-80 ms	50-80ms	Yes	N/A
CSIDH	512	10-20 ms	10-20ms	Yes	N/A

Conclusions

- What was discussed?
- What are the research results?